Table 2.1: Difference between light and electron microscope

S.No	Light microscope	Electron microscope
1.	Light is the illuminating source	The beam of electrons is the illuminating source
2.	Specimen preparation takes usually few minutes to hours. Live or dead specimen may be seen	Specimen preparation takes usually takes a few days. Only dead or dried specimen are seen
3.	Condenser, objective and eye pieces lenses are made up of glasses	All lenses are electromagnetic
4.	Specimen is stained by coloured dyes	Specimen is coated with heavy metals in order to reflect electrons
5.	It has low resolving power (0.25µm to 0.3 µm). It has a magnification of 500X to 1500X	It has high resolving power (0.001µm), about 250 times higher than light microscope. It has a magnification more than 100,000X
6.	Vacuum is not required	Vacuum is essential for its operation
7.	Image is seen by eyes by through ocular lens	Image is produced on fluorescent screen or photographic plate

examine objects. Electrons are considered as radiation with wavelength in the range 0.001–0.01 nm compared to 400–700 nm wavelength of visible light used in an optical microscope. Optical microscopes have a maximum magnification power of 1000, and resolution of 0.2 µm compared to resolving power of the electron microscope that can reach 1,000,000 times and resolution of 0.2 nm. Hence, electron microscopes deliver a more detailed and clear image compared to optical microscopes. Table 2.1 differentiate electron microscope from light microscope.



In 1924, a French scientist, Dr. De Broglie, showed that an electron beam behaved like waves and had a wavelength much shorter than the sizes of molecules and atom when accelerated.

Types of Electron Microscopes

- Transmission electron microscopes (TEM)
- Scanning electron microscopes (SEM)
- Scanning transmission electron microscopes (STEM)

The electron microscope was invented in 1931 by two German scientists, Ernst Ruska and Max Knoll. Ernst Ruska later received Nobel Prize for his work in 1986.

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